Assignment project |
AI/ML Developer

AI-Powered Development Cost & Timeline Prediction System

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Project objective

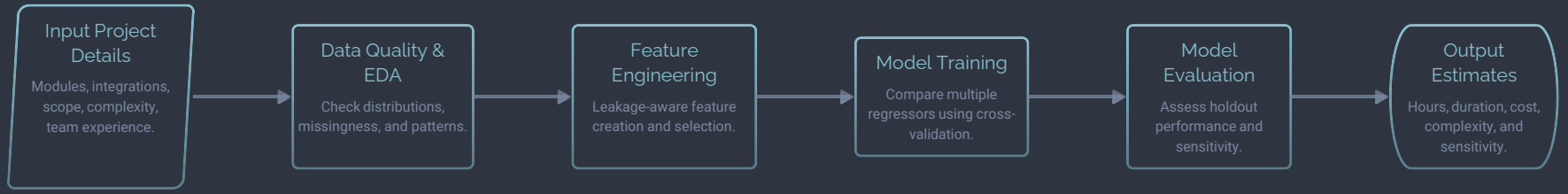
Build a working system that accepts project details and returns:

- ❑ Estimated development hours
- ❑ Estimated completion time in days
- ❑ Indicative cost in INR
- ❑ Complexity: Low, Medium, High, or Very High
- ❑ An explanation of which inputs influence the models overall

The problem

Software project estimates depend on requirements such as modules, integrations, feature scope, complexity, and team experience. Estimating these consistently at the planning stage can be difficult.

Estimator Pipeline



Estimator inputs and outputs

Project inputs

Project type, modules, user roles, integrations, feature requirements (Payment, Admin, Mobile, AI/ML), complexity, team size, and experience.

Project duration

Model prediction.

Complexity score

Rule-based score.



Development effort

Model prediction.

Indicative cost

Effort × ₹1,200/hour.

Feature importance

Global feature importance.

Dataset

Understanding the synthetic nature and assumptions of the training data

- 400 synthetic project records (real records were unavailable).
- Complexity labels generated via rule-based scoring.
- Cost derived from effort using ₹1,200/hour assumption.
- **Key Assumption:** Synthetic relationships are illustrative and do not represent verified company history.



Displaying first 5 rows:

	project_id	project_name	project_type	num_modules	num_user_roles	num_integrations	payment_gateway_required	admin_panel_required	mobile_app_required
0	PRJ-0001	Synthetic Project 0001	CRM	11	4.0	3	No	Yes	No
1	PRJ-0002	Synthetic Project 0002	Mobile Application	4	5.0	NaN	No	Yes	Yes
2	PRJ-0003	Synthetic Project 0003	E-commerce	4	4.0	NaN	No	No	No
3	PRJ-0004	Synthetic Project 0004	E-commerce	14	6.0	2	Yes	Yes	No
4	PRJ-0005	Synthetic Project 0005	E-commerce	15	4.0	1	Yes	Yes	No

Displaying first 5 rows:

ai_ml_required	ui_ux_complexity	backend_complexity	num_developers	developer_experience	development_hours	duration_days	development_cost_inr	complexity_level
No	High	Low	7	Mid-level	798	26	957600	Medium
No	Medium	Low	6	Mixed	519	26	622800	Medium
No	Medium	Low	3	Junior	409	24	490800	Medium
No	Low	Medium	8	Mid-level	713	18	855600	Medium
No	Medium	Medium	7	Senior	685	22	822000	Medium

Data quality is checked before modeling

Findings by issue type:

issue	count
missing_value	96
category_normalized	36
numeric_suffix_removed	9

Issues handled

Missing values, inconsistent formats, and numeric noise (141 logged injected issues).

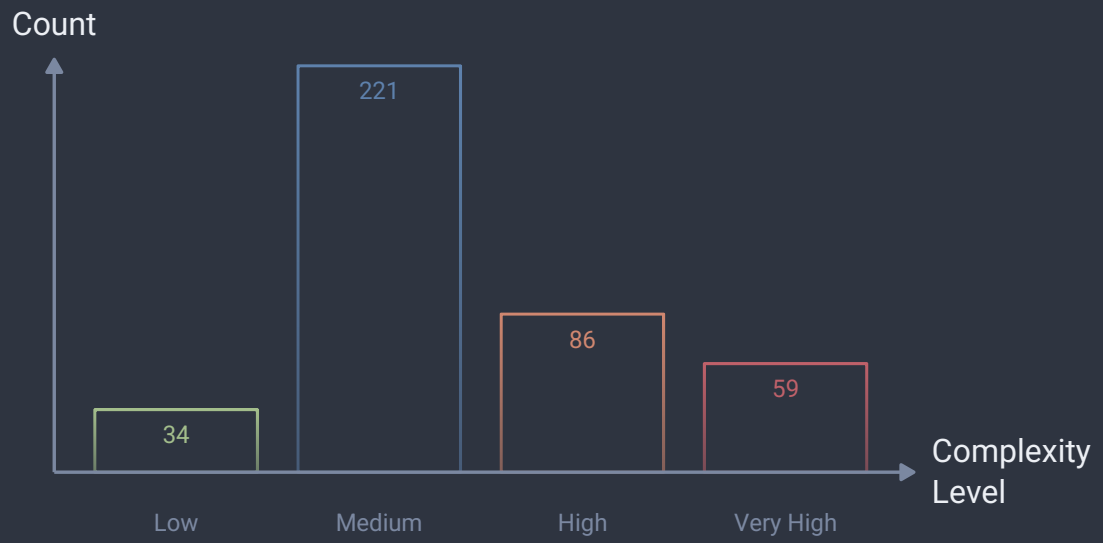
Workflow

Validate categories, check numeric ranges, consistent missing-value representation, and pipeline-integrated imputation/encoding.

Ensuring robust predictions through rigorous preprocessing

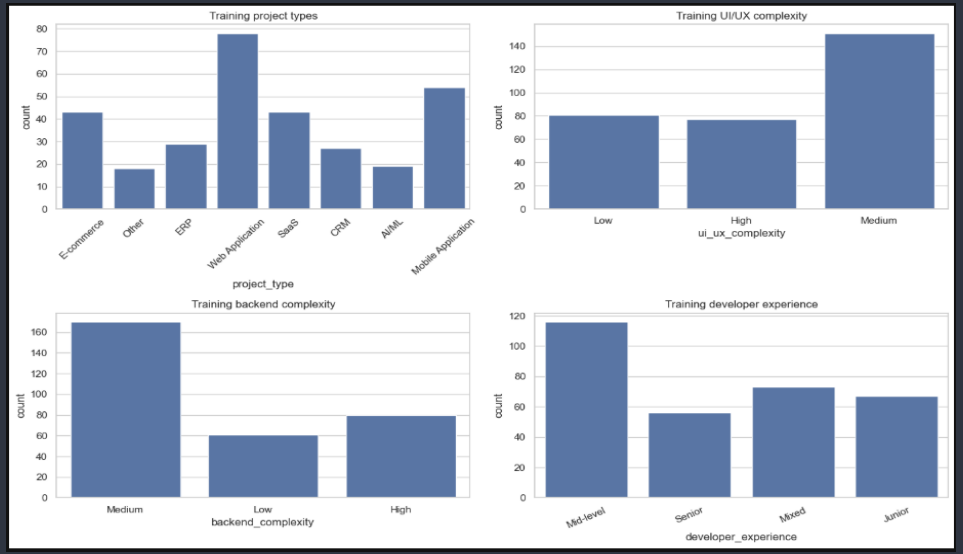
EDA checks distributions, missingness and patterns

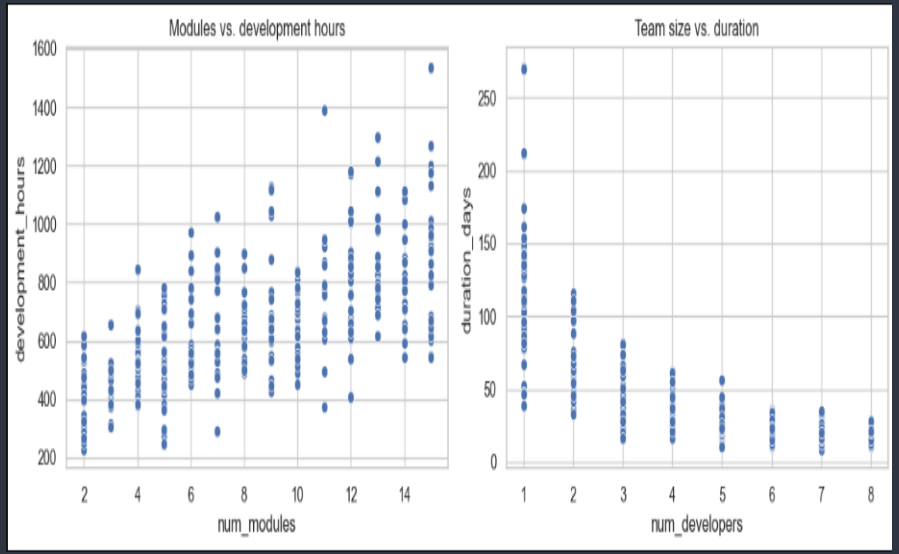
Analysis of development-hours and duration distributions. Investigation of relationships between project inputs and targets.

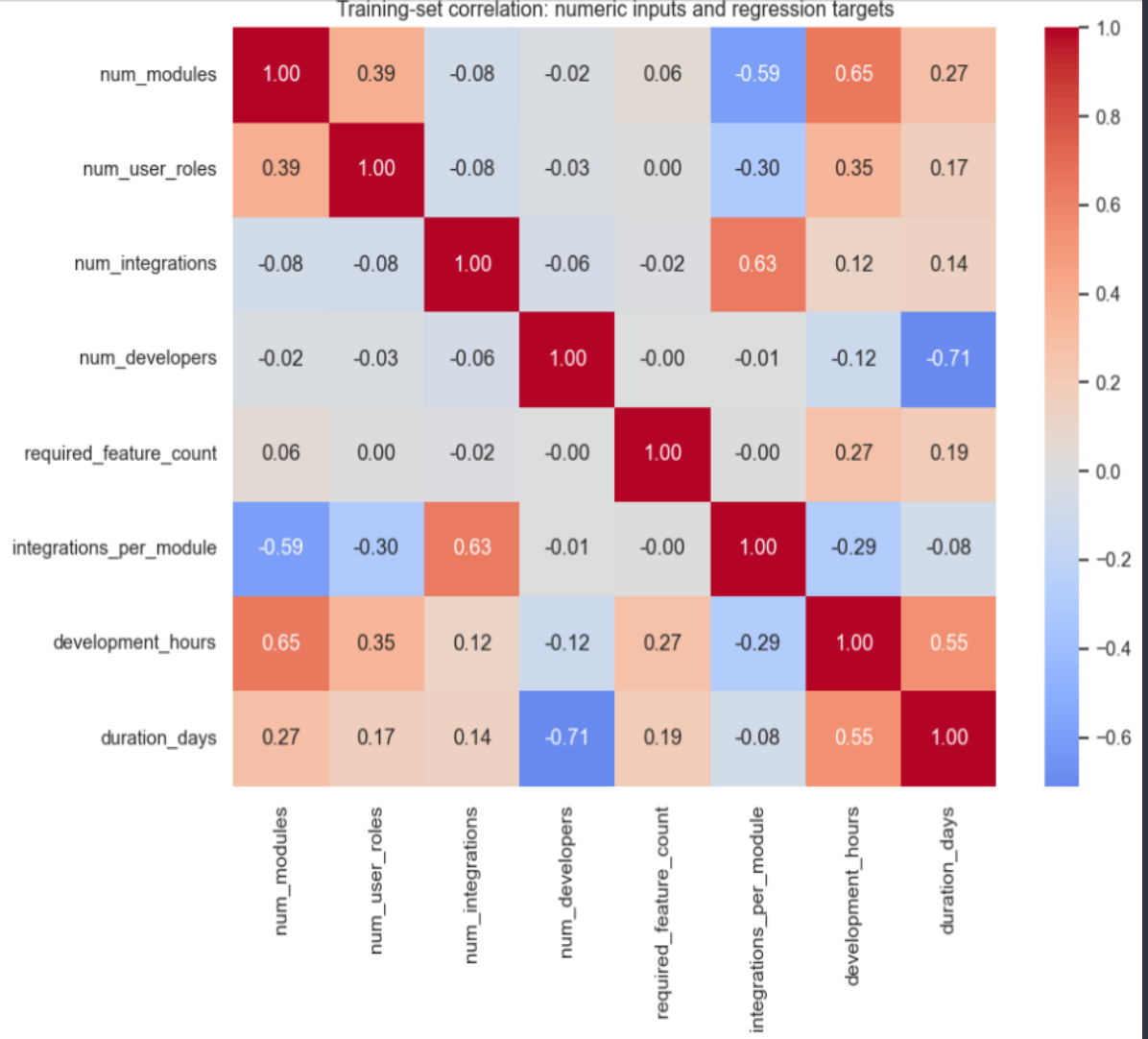


- Web applications are the most common project type in this training sample. Medium UI and backend complexity, and mid-level developer experience, are also the most frequent categories. These are sample distributions, not evidence of market demand or model importance.

- Projects with more modules generally show higher development hours. Duration tends to decrease as team size increases in this synthetic sample. These are observed patterns, not proof of causation; the model learns from the dataset's generated relationships.







Correlation heatmap:

Development hours have the strongest positive correlation with module count (0.65). Duration has a strong negative correlation with developer count (-0.71). These are patterns in the synthetic training data, not proof of causation.

Feature engineering and leakage-aware evaluation

01

Engineered features

required_feature_count, integrations_per_module.

02

Leakage controls

Targets excluded from inputs; holdout set untouched during selection; imputation/encoding fitted within training folds.

03

Model selection

Model selection uses training cross-validation only.

Multiple regressors are compared with cross-validation

Selecting the best-performing models for effort and duration



Models compared

Linear Regression, Random Forest Regressor, XGBoost Regressor.

Evaluation

Repeated 5-fold cross-validation on training data.

Metrics

MAE (average error), RMSE (penalizes large errors), and R^2 (goodness of fit).

Selected models and holdout performance

Note: Linear Regression was selected for hours based on training CV

	target	model	CV_MAE_mean	CV_RMSE_mean	CV_R2_mean
1	development_hours	Linear Regression	97.091005	127.298661	0.643410
3	development_hours	XGBoost	105.775925	140.255475	0.570214
2	development_hours	Random Forest	113.608523	149.027665	0.515662
0	development_hours	Dummy Median Baseline	168.436875	217.357139	-0.027966
7	duration_days	XGBoost	10.103378	18.007028	0.778797
6	duration_days	Random Forest	10.401484	17.604826	0.784404
5	duration_days	Linear Regression	17.493027	25.294080	0.561677
4	duration_days	Dummy Median Baseline	23.979375	40.779481	-0.129370

Target Metric	Development hours	Duration
Model	Linear Regression	XGBoost Regressor
MAE	97.26h	8.58d
RMSE	144.07h	14.34d
R^2	0.649	0.889

Separate training from live prediction



Offline

Data checks -> Engineering -> Pipeline -> CV/Selection -> Saved Models.



Live

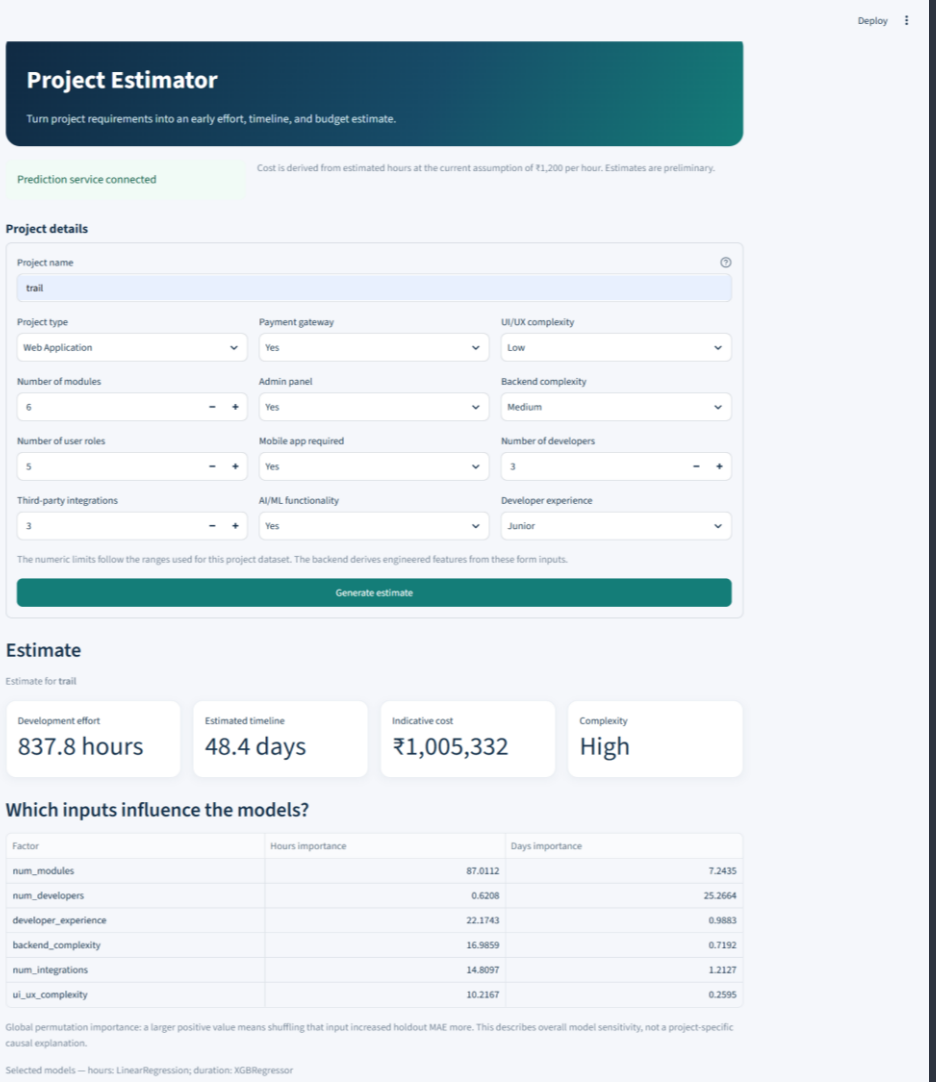
Streamlit UI -> FastAPI Validation -> Feature Derivation -> Model Estimation -> Dashboard.



Endpoints

/health, /model-info, /predict.

A modular architecture for scalable project estimation



Live Estimator Demo

For the sample project, the app estimates 837.8 hours, 48.4 days, ₹1,005,332, and High complexity.
The importance table shows module count as the strongest overall influence on hours, and developer count on duration. These importance values describe the models overall, not this specific estimate.
Note that cost uses the unrounded hours prediction

Working prototype; real-world validation is the next step

The current deliverable demonstrates a complete, leakage-aware ML workflow and a working application.

